STRATUM

Predicting and avoiding persistent contrails over Indian airspace

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STAGE	VERSION	DATE	TAGLINE
Activation (concept)	1.0	12 October 2026	Move the plane. Lose the cloud.

1 Summary

When a jet flies through cold, humid air, the water vapour in its exhaust can freeze into a thin line of ice cloud: a contrail. Most fade within minutes. In air that is already supersaturated with respect to ice, they spread into cirrus sheets that last for hours and trap heat that would otherwise leave the Earth.

The best current estimates put contrail cirrus at roughly half of aviation's warming effect, which is more than the CO_2 from the fuel itself, though the uncertainty is wide.[1,2] Unlike CO_2, a contrail can often be avoided by flying a couple of thousand feet higher or lower, and a small share of flights produces most of the effect.[2,3] In a randomised trial with a US airline, flights that adjusted altitude made 64% fewer contrails for about 2% more fuel on each adjusted flight.[4]

The hard part is prediction. Weather models are weak on humidity at cruise altitude, so contrail forecasts are wrong often enough that dispatchers do not trust them.[5,6] The research and trials that exist are concentrated over Europe, the North Atlantic and the United States. Over India, the world's third-largest domestic aviation market,[9] we have not found any study that checks contrail forecasts against what satellites actually see.

STRATUM is a system we are designing to make that check and then fix what it finds. It combines open flight tracks, ERA5 weather data and Meteosat satellite imagery to predict which flights over India will form persistent contrails, learn from satellite-observed contrails where the humidity forecast goes wrong, and give each flagged flight an altitude alternative with an estimate of the warming avoided against the extra fuel burned. This synopsis covers the problem, our objectives and method, what is new about the approach, and what we plan to deliver by April 2027. STRATUM is at the concept stage, so no results are reported here.

2 Background

2.1 Why contrails matter

Aviation affects climate in several ways. CO_2 is the one everybody knows, and it is the one airlines report. The largest single term in the most widely used assessment is contrail cirrus, at 57.4 of the 100.9 mW m^{-2} net effective radiative forcing from aviation in 2018, against 34.3 mW m^{-2} for CO_2.[1] A later global study using flight-by-flight modelling put the 2019 contrail forcing about 44% lower than the earlier best estimate, with a range of 34.8 to 74.8 mW m^{-2}.[2] Contrails stay a large term in either estimate. What the revision shows is how much depends on the humidity inputs, which is the problem STRATUM is built around.

Three facts make contrails a practical target:

FINDING	WHAT IT MEANS FOR AVOIDANCE	SOURCE
About 2% of flights caused 80% of global contrail energy forcing in 2019	We do not need to move every aircraft. Finding the few that matter is the whole job.	Teoh et al. 2024[2]
Night flights were about 25% of traffic but 60–80% of contrail forcing in a UK study	At night a contrail only traps heat; there is no sunlight for it to reflect. Overnight routes, many of them cargo, deserve priority.	Stuber et al. 2006[7]
Treatment flights in a randomised airline trial had 64% fewer contrails for about 2% more fuel per adjusted flight	Altitude changes work in real operations, not only in simulations.	Sonabend-W et al. 2024[4]

2.2 Why India

India is now the third-largest domestic civil aviation market. Its commercial fleet grew from 395 aircraft in 2014 to 844 in 2025, and domestic traffic has grown about 9% a year over the last decade.[9] Almost all contrail research, and every avoidance trial we know of, is over other regions.

The one global estimate that covers India gives a surprising answer. It found that India and China together account for 19% of flight distance but only 6% of contrail energy forcing, and attributed this to the tropical circulation and lower cruise altitudes.[2] That may be right. But the same study notes that the humidity correction is the most influential input, and that correction was tuned on in-situ aircraft measurements collected mainly elsewhere. ERA5 is known to have a dry bias near cruise altitude that makes it underpredict persistent-contrail conditions.[6] So there are two possibilities. Either India's contrails really are a smaller problem per kilometre flown, or they are being underestimated because nobody has checked. Both answers are worth having, and with the fleet doubling, the absolute amount will grow either way.

3 Problem statement

> *Airlines and air navigation providers over India have no validated way to know, before departure, which flights will form warming contrails, how confident that prediction is, and what avoiding them would cost in fuel.*

4 Aim and objectives

Aim. Build and test an open, satellite-checked contrail prediction system for Indian airspace that ranks flights by how much warming could be avoided and at what fuel cost.

NO.	OBJECTIVE	WE WILL KNOW IT IS DONE WHEN
01	Build a flight-level contrail formation and persistence model for Indian airspace from open data.	Every cruise segment on a chosen corridor gets a formation flag, a persistence flag and a modelled lifetime.
02	Measure how well raw ERA5 humidity predicts contrails that satellites observe over India.	We report hit rate and false alarm rate against a labelled satellite set, with confidence intervals.
03	Train a correction model on satellite observations that accounts for the contrails satellites cannot see.	The corrected model is scored on months it never trained on and compared with published correction methods.

NO.	OBJECTIVE	WE WILL KNOW IT IS DONE WHEN
04	For each flagged flight, compute altitude alternatives and the warming-versus-fuel tradeoff.	Each flight in the ranked list shows the alternative flight level, extra fuel in kg, and warming avoided as a range.
05	Publish a free web dashboard and an open dataset.	Anyone can open the map, see the method, and download the labelled data.

5 Research questions

1. **RQ1.** How well does raw ERA5 humidity, run through standard contrail physics, predict the persistent contrails that satellites observe over India?
2. **RQ2.** Can a model trained on satellite observations correct those predictions, and by how much, on months it has never seen?
3. **RQ3.** How concentrated is contrail warming over India across flights, routes, flight levels and time of day, and what would avoiding the worst cases cost in fuel?

6 Method overview

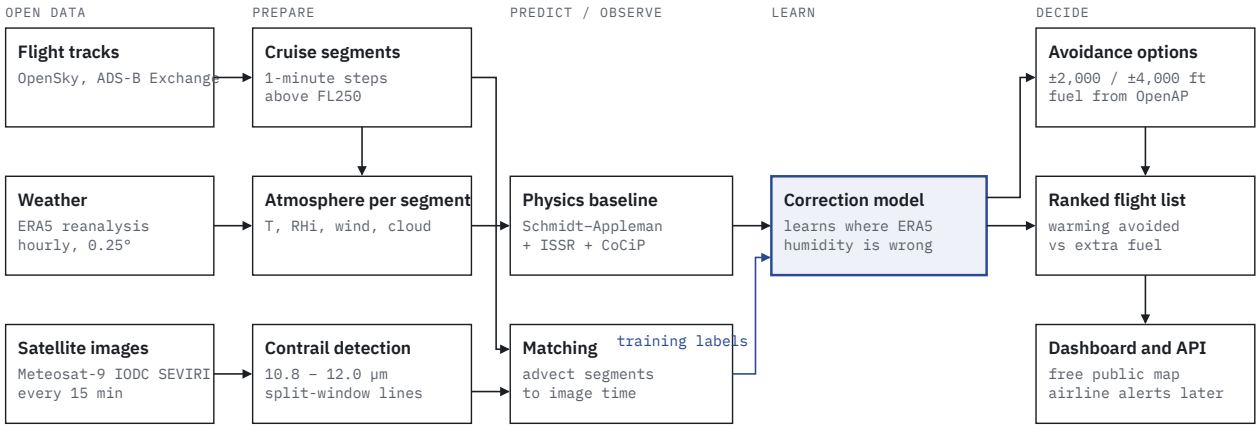


Figure 1. The STRATUM pipeline. All three inputs are open data. The step that makes STRATUM different is the correction model (highlighted), which is trained on contrails that were actually observed by satellite.

Flight tracks. We take ADS-B positions from the OpenSky Network[12] and the free monthly sample days published by ADS-B Exchange, and cut each flight into one-minute cruise segments above FL250. OpenSky's receivers are concentrated in Europe and the US, so our first task is an audit of how much Indian traffic each source actually covers.

Physics baseline. For every segment we read temperature, humidity and wind from ERA5[10] and apply the Schmidt–Appleman criterion to decide whether a contrail forms. If the air is supersaturated with respect to ice, the contrail can persist, and we run the Contrail Cirrus Prediction Model (CoCiP)[11] through the open-source pycontrails library to estimate its lifetime and energy forcing.

Satellite observations. Meteosat-9 sits over the Indian Ocean at 45.5°E and images the region every 15 minutes. Contrails show up as thin lines in the difference between its 10.8 and 12.0 µm channels. We will detect them automatically and hand-label a validation set, with two team members labelling each scene so we can measure how much we agree.

Correction. We move each flight segment forward with the ERA5 winds to where its contrail would be at image time, then compare prediction with observation. A model learns, from features such as humidity, temperature, vertical humidity gradient, season and time of day, where ERA5 is too dry or too wet. Satellites miss many thin contrails; one study found that even an idealised 2 km imager would see only about 46% of contrail segments.[8] We therefore train only on scenes where a contrail could have been seen, and treat the rest as unknown, not as "no contrail".

Decision output. For each flagged segment we test ±2,000 and ±4,000 ft alternatives, estimate the extra fuel with the open aircraft performance model OpenAP, and report warming avoided as a range. The dashboard shows a ranked list, not just a map.

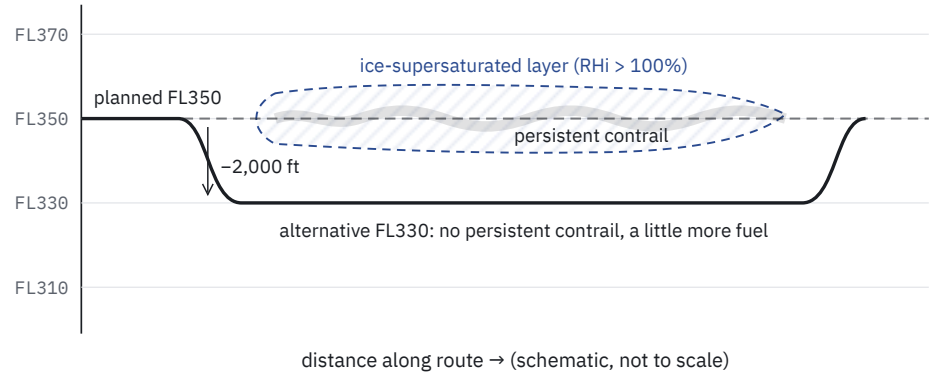


Figure 2. The kind of decision STRATUM supports. Ice-supersaturated layers are usually shallow, so a small altitude change can take a flight out of one.

7 What is new

- **The region.** We have not found a published study that validates contrail forecasts over Indian airspace against satellite observations. Global estimates for India rest on corrections tuned elsewhere.
- **Learning from what was seen, with the blind spots modelled.** Most correction methods are fitted to in-situ humidity measurements. We fit to observed contrails, which are the thing an airline cares about, and we treat unobservable scenes explicitly.
- **A decision, not a map.** Each alert says which flight, which alternative level, how much extra fuel and how much warming avoided, with uncertainty shown as a range.
- **Open by default.** Code, labelled data and evaluation results will be public, including the experiments that do not work.

8 Deliverables

DELIVERABLE	TARGET DATE
Lean Canvas (Activation Stage)	29 Oct 2026
Data coverage audit: how much Indian cruise traffic OpenSky and ADS-B Exchange capture	22 Nov 2026
Prototype v0: physics-only contrail map for the Delhi–Mumbai corridor, one month	20 Dec 2026
Innovation Brief, Innovation Video and website (Innovation Stage)	7 Jan 2027
Labelled satellite contrail set (200 scenes) and baseline evaluation	28 Feb 2027
Correction model and dashboard v1	April 2027
Preprint and open dataset release	mid-2027

9 Scope and limitations

In scope

- Commercial jet traffic at cruise over Indian airspace
- Persistent contrails and their modelled warming
- Altitude changes only, not lateral reroutes
- Offline analysis and next-day forecasts from archived data

Out of scope for now

- Live operational use by airlines or air traffic control
- Airspace capacity and ATC conflicts from altitude changes
- Non-CO_2 effects other than contrails, such as NO_x
- Any field trial with real aircraft

The main limitations are known in advance. ADS-B coverage over India may be patchy. Satellites see only a subset of contrails, and monsoon cloud will hide many more for part of the year. ERA5's 0.25° grid is coarse compared with thin humidity layers. The size of the warming from a contrail is itself scientifically uncertain, which is why we report ranges. All of our work is computational.

10 Team

MEMBER	ROLE (PROPOSED)	MAIN RESPONSIBILITIES
Shrugal Vinod	Modelling lead	Correction model, evaluation design, pre-registered hypotheses
Aditya Harsoda	Flight data lead	ADS-B ingestion, trajectory cleaning, coverage audit, fuel estimates
Ishaan Kupnur	Satellite lead	Meteosat processing, contrail detection, labelling protocol
Abeer Shah	Product and business lead	Dashboard, website, video, market research, outreach
Dr. Varsha Pant	Supervisor	Scientific review, milestone checkpoints, advisor introductions

11 Timeline

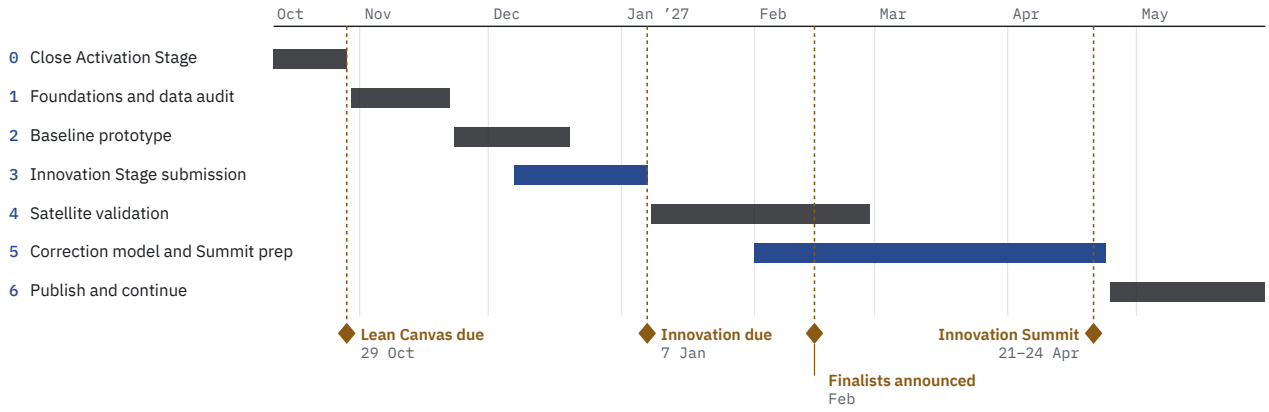


Figure 3. Project phases against the Conrad Challenge deadlines. Phases 3 and 5 (blue) are the competition submissions. A full task-level plan is in the separate phase-by-phase plan.

12 Resources and budget

All the data STRATUM needs is free for research use: ERA5 through the Copernicus Climate Data Store, Meteosat imagery through the EUMETSAT Data Store, and flight tracks from OpenSky and ADS-B Exchange. The software is open source (Python, xarray, pycontrails, OpenAP). Compute for a single corridor fits in free tiers such as Google Colab and Kaggle.

ITEM	ESTIMATE	NOTE
Conrad Innovation Stage entry fee	US$499	Financial aid can be requested
Website domain, one year	≈ ₹1,000	Hosting on a free static host
Data, software, compute	₹0	Free research access and free tiers
Poster and printed material if selected as finalists	≈ ₹3,000	Innovation Summit expo

13 Use of AI tools

We used an AI assistant while preparing this synopsis, mainly to find and summarise papers, to explain parts of the contrail physics, and to suggest edits to our drafts. Every reference was opened and checked by a team member, and the project design and decisions are our own.

– References

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